

NAG STRATEGIC BACKGROUND

Highly Confidential to NAG and advisors – not fit for distribution

1 DOCUMENT PURPOSE

A brief, imprecise overview of the strategic background in which NAG is operating. This is not meant to provide a strategic analysis but may facilitate in the development of one. This document can be read by NAG Strategic advisors to get a sense of what the company is currently doing before real discussions can begin.

2 HISTORY

NAG has existed in some form since 1970 and has been a commercial company since 1976. The NAG Library, a collection of numerical algorithms, has been the primary product and revenue generator through this period and remains one of two major revenues streams today. Through the 1980s and 1990s NAG developed a strong reputation in the area of numerical analysis and numerical software development. The product became world-recognised, both in the user community and in the academic research community. Major academic / research players contributed to the NAG Library in areas such as Statistics, Linear Algebra and Mathematical Optimization. NAG remains strongly linked to world class researchers, such as Jack Dongarra (Manchester & Tennessee - HPC, Linear Algebra) Nick Higham (Manchester - numerical Analysis) and Uwe Naumann (RWTH Aachen, Automatic Differentiation) to name just a few.

The NAG Library remained the strongest collection of algorithms through a competitive period, with IMSL (Visual Numerics, now Perforce) the closest direct competitor, having a stronger presence in the US but weaker in the UK and EMEA. Students were taught the NAG library in their Fortran and engineering undergraduate courses and entered the workforce with knowledge of the product and its abilities. It remained well-recognised and well regarded name in the field of numerical analysis and HPC and was used in a variety of Engineering sectors.

Over the period 2000-2010 the NAG Library developed a strong customer base in sell-side Financial Services, as the investment banks that developed their own pricing libraries required a robust and supported algorithmic underpinning to their own codes. NAG today remains strongly embedded in a majority of the tier-1 banks and several smaller banks. The financial sector is a rather impenetrable sector in which the customer's technical problems are not always articulated well for proprietary/competitive reasons. NAG has been unable to either strongly tune the product to the needs of these customers, nor to capitalize on the market penetration by developing successor products (with the exception of the AD Suite, mentioned below). The financial clients are mostly on perpetual licenses with limited annual service/support revenues for NAG.

During a similar period, the fundamental algorithms that affect processor performance (linear Algebra, FFTs, RNGs) started being developed in-house by processor vendors to provide value-add to their products. These companies could exploit more specific tunings for their own hardware and as such NAG could not develop

products that were competitive. This part of the NAG library's business was lost, although NAG continues to partner in the development of those libraries (e.g. Arm, Intel).

The Library business model has some natural limitations, it is a collection of user-agnostic functionality that covers a broad mathematical spectrum. As such, traditional product management techniques that tailor the product's feature sets to a perceived need in the market, are difficult to apply. Content has sometimes been added to the library according to the interests of the research community or opportunistic developments inside the company, with little commercial awareness. Recent attempts to address this have been fraught with obstacles. A trio of conditions make this problem difficult – the user-agnostic nature of numerical content, the aggregation of content into the library format, and the proprietary nature of many of NAG's clients.

NAG has over years attempted to develop or productise various other technologies, such as new compilers, visualization frameworks and PDE solvers (twice). None of the new products has ever had any strong success. However, the GenStat product that was spun-off to start a new company VSN has been mildly successful, and NAG still owns a 25% stake in that company.

NAG has historically performed technical services alongside numerical library development, providing both an alternative (minor) revenue stream and boosting library renewals. Major service work was established with technology providers such as Intel, Arm and AMD, usually to develop some portion of their numerical software. NAG developed the bulk of AMD's ACML library in the early Opteron period. In the 2000s, NAG began providing HPC services as an offering and established a major contract with HECToR, the UK national resource, to provide CSE support. For approx. 7 years (2007-2014) NAG successfully established this service and the company became both larger and financially very strong on the back of this. NAG was aggressively outbid in the HECToR follow-on project called ARCHER, lost the business, and as a result went through significant reductions in staff.

Around 2010 NAG developed a relationship with an O&G major in Houston, through which a modest amount of HPC service work ensued. Over a number of years NAG developed the relationship to the point of providing continued HPC service provision on behalf of the client, a continual arrangement with >10 staff provided for many years. This relationship has provided a sustainable revenue stream with strong margins for several years. NAG has failed to replicate the model, which seems to be strongly tailored to the clients' internal difficulties recruiting technical staff.

NAG has provided a commercial Fortran compiler since the 1980s. It is world renowned in a niche setting – strong support of the Fortran standard and error checking. It is used by many Fortran customers as a second, supporting compiler to help development. It has never been a commercially lucrative product but provides some reputational value to NAG.

NAG has commercialized the Automatic Differentiation products of RWTH Aachen. These products have found a market within Financial Services. The company has reason to believe that automatic differentiation (similar to back-propagation in Machine Learning) represents some real opportunity for NAG but this is yet to be realized. The collaboration with RWTH Aachen is extremely strong and has been recently ramped up, NAG funds multiple PhD positions there, with those students working part-time as engineers at NAG. The cost of sale of current the product is high, revenues of the product remain low and penetration outside of FSI is low.

Brian Ford was NAG's founder and CEO until 2005, Rob Meyer was CEO in the period 2005-2019 and Adrian Tate is the current CEO. Since 2019, NAG has been undergoing a period of transformation, with changes to the Executive team, collaboration style, development practices, emphasis on product, and marketing. The company structure and culture have been drastically altered. Progress has been made in terms of innovation and sustainability but many of NAG's problems remain. In particular, the company is considered to have a lack of focus, with too many balls in the air, too little time spent engaging with the market and too long spent talking internally with itself.

3 CURRENT PRODUCT PORTFOLIO AND STRENGTHS

NAG LIBRARY

Revenues (GBP) : Approx 4 Million GBP

Profitability : Low (pre-2022), High (post-2022)

Value Proposition : Robust, supported collection of algorithms,

Key Clients : Most banks, some major industrial (Airbus, NeilsenIQ,

Competitive Position : Owns the market in general libraries. Budget option version the Optimization Libraries (Gurobi,

Strengths : High barrier to exit, solution gets baked into applications and sticky, renewals strong, attrition low

Weakness : Cost of Sale is high, new business very low and decreasing

Trends : Value is eroding over time as new usage models become dominant, new business very low, not able to corner any of the data-science boom

MANAGED SERVICES

Revenues : Approx 4 Million GBP

Profitability : medium

Value Proposition : Avoids internal recruiting concerns via outsourcing of HPC service

Key Clients : O&G major Client (only one)

Competitive Position : Strongly embedded in the client, major competition is the decision to do in-house, currently not likely,

Strengths : current relationship remains strong, but personnel changes will risk this

Weakness : considered to be “not replicable”, also not something that can be discussed openly

Trends : Losing people to the “great resignation”

HPC CLOUD SERVICES

Revenues : Approx 1 Million GBP

Profitability : zero

Value Proposition : get better cost-to-solution from your migration to Cloud HPC with mix of traditional HPC tuning and Cloud / sysadmin support

Key Clients : via the CSPs - Google, Microsoft, AWS

Competitive Position : Red Oak and other UK small business compete for the business, main competitor is the client organisations doing the work in-house

Strengths : momentum achieved through the GTM efforts of the last year, strong / building relationship all three major CSPs

Weakness : business has been patchy and slow, but picking up

Trends : Cloud is growing, we are hitting the tipping point for HPC but clients still struggling

AUTOMATIC DIFFERENTIATION SUITE

Revenues : Approx. 0.5 Million

Profitability : zero

Value Proposition : calculating sensitivities much more quickly can save your business lots of money

Key Clients : tier1 and tier2 investment banks, some automotive and F1

Competitive Position : open-source software exists that is comparable, but very hard to use. Startups MatLogica and MOCAx have strong niche products and aggressively marketed.

Strengths : great relationship with RWTH Aachen has enabled and resourced the product development. This is really a solutions space, product and service together achieve results, we have an advantage in this regard compared to small players.

Weakness : Our pricing appears to be wrong, but we already have perpetual licenses with major clients. Poor traction outside of FSI.

Trends : the need for sensitivities happens everywhere that volatility happens, and we expect it to grow. Scientific domains and machine learning are two areas of interest.

NAG FORTRAN COMPILER

Revenues : Approx. 0.3 Million

Profitability : zero

Value Proposition : you need a second compiler to do error-checking and standard support properly.

Key Clients : US national labs

Competitive Position : NAG has a niche product that serves its niche well with zero growth potential.

Strengths : World-recognized in a very small niche, supports Co-arrays and more of the standard than other compilers

Weakness : No optimization capability, not for performance

Trends : Fortran continues its resurgence and will be embedded in performance codes for many years/decades. The world has moved onto LLVM-based compilers and all collaboration requires this basis.

In March 2022 NAG entered into a Framework Agreement with AMD that allowed AMD to make offers of employment to NAG employees. For each employee that accepted the offer, NAG would be rewarded financially. At the time of writing the transaction is yet to complete but is set to be quite successful. This style of transaction is related to “acquihire” (where a company is acquired for its staff) which is a recent trend and a reaction to the Great Resignation / global talent crunch. The scope of the deal means it is expected to drastically accelerate the reprogramming of NAGs internal dynamics and will increase profitability of the NAG Library. It is expected that the replacements to departing staff will be at two levels a) junior staff to build the next generation of engineers b) technical staff with more domain knowledge on how our products are used in the market, to drive innovations.

The people that have moved to AMD have done so voluntarily, and with an enormous increase in benefits. The transaction is much more amicable and positive than M&A activity.

Following the transaction, plus the existing healthy cash position, NAG has a cash balance of around 18 Million GBP, which is expected to be used to invest in meet strategic aims. This should be a mixture of rebuilding to support organic growth and inorganic activity to accelerate both growth and innovations. For those staff that did not move, this growth and focus period will represent an exciting opportunity for the company.

We feel that our supplies of UK based technical talent remain open and that we could restaff any new enterprise rather quickly. This leads to the potential for further Acquihire transactions with AMD or with other large technology providers.

5 NAG STRATEGIC AMBITION

NAG’s current strategic path has been around transformation. We have built missing capabilities and completely reprogrammed the internal dynamics of the company. In an older company, change can happen too quickly so this has been paced according to what the company can handle. We have built an innovation system, taken new services to market, developed both Product Management and Product Marketing functions, started new product discovery and new market discovery. Despite these successes, NAG is still too internal focused, struggles with the market information needed to make product decisions, and is yet to display real success outside the core areas of numerical software and managed services.

The NAG brand has been neglected over the years, and while there are some sectors that NAG has strong reputation for its size, there are other sectors where the name is not well regarded. The company still looks old-fashioned. Further improvements in presentation must be made, but the purpose of the company and the way that it positions itself to the outside needs a reset. It will be important to not throw the baby out with the bathwater.

NAG is primarily a products/technology company and will innovate to develop new IP in key areas. Candidate areas are HPC technology in Cloud enablement, Optimization software targeting Portfolio Optimization, and new AD products with a higher-level of abstraction. These product developments will be expensive, high-risk and potentially dangerous for a company of this level. Ways to de-risk these efforts are central to NAG’s sustainability plans.

One of NAG’s traditional strengths has been the cross-over of services and product development, meaning that inefficiencies on both sides can be mitigated by staff moving between them. The recent success with acquihire transactions means that NAG may have an interesting and unique path to further de-risking product offerings, by including the option of such transactions in the paths to exit. New service offerings can be very difficult to make sufficient margins due to consultant utilization, and new product offerings are naturally risky due to product-market fit. NAG can explore acquihire transactions simultaneously to these other enterprises, which means that when developments or new offerings do not become sufficiently profitable, then another route to

profitability can be taken. NAG may have a unique ability to consider the Acquire transaction as a strategic route, due to the availability of RSE-level talent in the UK and the attractiveness of NAG as employer to those semi-academic staff.



NAG's status as a not-for-profit group is useful for these purposes, and for maintaining the collaborations with industry that allow NAG to punch above its weight. However, NAG intends to start for-profit subsidiaries inside the group, through which equity participation and outside investment are legitimate considerations. "NAG Solutions" is expected to be incorporated this year, with benefit structure more appealing to a new generation of technical. NAG Solutions can be used as an exit vehicle for acquire transactions.